

Pathways to Equality: Mapping Gender Wage Equity by State and Industry

Ronald Sowadski

Business Consultant, Grand Canyon
University, Arizona, United States

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***Corresponding Author :Ronald Sowadski**

Email : RonnieC312@gmail.com

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Abstract :

The cause of this examine became to study whether or not there are variations in gender-related pay equity in the public quarter, amongst states, and among industries inside the United States. The study changed into carried out with archival statistics from the American Community Survey. Results of one-manner ANOVAs showed a substantial distinction in gender-related pay equity a number of the 51 states (which include D.C.), $F(50, 1740) = 1.69$, $p = 0.019$ and most of the 5 foremost industries, $F(4, 1735) = 17.00$, $p < 0.01$. These empirical findings offer a basis for the improvement of rules needed to deal with pay inequity.

Keywords : Hemoglobin, Hematological disorders, DNA sequencing, HBB gene

Introduction :

Up to date updated 1964, pay fairness has been a social difficulty that the government, the entity accountable for the regulation of pay, has now not been up to date updated solve. this is due-in element-updated the lack of an entire dataset [1]. This take a look at and the trouble of pay equity are up-to-date on fairness principle, truthful distribution of contributions, and blessings for absolutely everyone [2]. previous pupils have utilized fairness principle updated pay equity [3-5]. diverse legal guidelines and acts were implemented; however, little development has been made with this social trouble. irrespective of a couple of federal laws implemented, women nonetheless earn less than men [6]. over the last century, the gender-related pay equity trouble has narrowed, however it remains tremendous [7]. The federal government has stated that the pay equality gap is 20% and could take another a hundred thirty years up to date solve [8, 7, 4]. This have a look at my additionally disclose what states and industries make contributions updated the social trouble of inequity and provide proof which can provoke alternate.

The body of studies on pay equity has not unusual issues, which includes laws, acts of rules, government adjustments, societal focus, and human useful resource departments' responsibilities [9, 3, 5,10]. destiny studies require an assessment of why the gender-related pay fairness problem has now not been researched extra very well. that is so in spite of the Pay Equity Pay Act being exceeded in 1963. at the present day charge, the gender-associated pay equality issue will now not be closed until the year 2152 [11, 7, 4]. This look at's findings can also result in changed rules at all tiers of government and society, thereby improving the danger of obesity, heart attacks, depression, and social/monetary inequity of seventy 4.6 million women. Extra medical know-how of pay equity is needed. This observe's blessings may additionally provide societal consequences which could exchange the U.S. financial system whilst enhancing the field of commercial-organizational psychology. an extra 512.6 billion dollars might be coming inupupdated the USA economy if women were paid equally as men [12]. .

The USA gross national profit is nineteen.61 trillion dollars and offers almost a 3% growth inside the united states financial system. A 512.6 billion dollars consistent with year's inflow in updated the financial system could positively trade American society as a whole.

The negative outcomes effects of the gender-related pay equity difficulty are felt for the duration of the economic system and society. Pay fairness is a subject updated that influences 74.6 million girls people in the civilian hard work force [13]. Same pay between males and females might reduce poverty for running women from 8.2 updated 4%. Each of the 50 man or woman states might gain from an increase of funds in up-to-date their economies [14]. The most substantial damaging effect of gender wage inequality is that pay fairness problems contribute up-to-date increased anxiety and depression quotes amongst ladies [15].

In a partnership with the center for place of work intellectual fitness, the yank Psychiatric association expressed great situation on up-to-date of gender-related pay fairness. The President of the American Psychiatric affiliation stated that gender-associated pay fairness troubles are extra impactful than financial troubles and contribute up to date mental problems [16]. Moreover, more recent researchers have proven that income inequality increases the danger of obesity and coronary heart attack [17]. It is truthful up to date updated that the detrimental outcomes of gender-related pay equity are sizable updated American society.

Previous researchers reading this updated have used restrained and small pattern sizes, which required extra research [18]. Researched pay fairness with a sample length of best eight of the 50 states. Moreover, these authors centered on 139 establishments, representing a restrained pattern length in the 8 states [19]. Researched pay equity, locating the observer's maximum large quandary updated be pattern size and the need up-to-date enterprise records. The sample length handiest covered 25 USA up to date, no longer representing all 50 states or entire states, and did no longer don't forget updated industry information [20]. Diagnosed an obstacle of his pay fairness research, which protected 23 out of the 50 states of the united states and simplest included metropolitan areas, not the entire kingdom [21]. Researched the gender-related pay equity trouble using a sample size that simplest consisted of less than 25,000 contributors, now not representing all 50 states, and requested a wholly nationally consultant dataset [4]. Researched gender-related pay equity problems tendencies but diagnosed limitations in her qualitative have a look at.

Rosado requested future research up-to-date provide a larger sample size, enterprise evaluation, and the use of a quantitative research method. The previous literature gap of restrained sample size has up-to-date the need for a qualitative nationwide records set that includes all 51 states (updated D.C.) and enterprise evaluation.

Additional gaps had been located inside the previous literature. Previous literature applied old sample records, which required in addition research. Many preceding studies that had barriers of sample length additionally utilized previous datasets [19]. Researched pay fairness and used a dataset that up-to-date eight years old at completion updated. The dataset that was used up to date updated the 2011 American network Survey statistics [21]. Gender earning fairness take a look at applied a 2010 dataset that became seven years old at up to date updated and requested future studies up-to-date profession and industries [20]. Researched gender earning fairness using a 17-yr-antique dataset from the year 2000. Within the current study, the researcher used the maximum most up-to-date dataset up to date make certain the validity of the outcomes.

The U.S. Census Bureau and American Network Survey facts is the most appropriate dataset for assessing country and industry-degree statistics. Information is given from a national perspective up to date cowl individual states and industries. This examine might also provide gender-related pay fairness information with the aid of nation and industry, with delivered analyses of capacity social and financial up-to-date contributing updated pay fairness troubles.

Materials and methods:

The purpose of this quantitative, non-experimental, comparative observe become up to date examine variations in gender-associated pay equity across states and industries inside the U.S. economic system's public region.

The have a look at become performed the use of archival records from the 2017 American network Survey. Out of 3,526,808 responses up to date that survey, simplest 2, a hundred 45,639 had been retained within the final dataset up-to-date the us Census Bureau best accepts completely completed surveys.

The researcher downloaded survey facts aggregated on the subindustry stage by using kingdom. The dataset analyzed in this examine protected 1,834 data up-to-date representing 36 subindustries X 51 states (including D.C.).

This explains why up to date wide variety of instances in the records record was 1,834 up-to-date 1,836. There were 36 subindustries within each nation, except the District of Columbia, which had most effective 34 subindustries. Subindustry information points exceptional represented up to population.

The impartial variables have been kingdom and enterprise and compared 1,834 records up-to-date for 36 subindustries inside every of the 51 states (up-to-date D.C.). The established variable up-to-date gender-related pay fairness, operationalized as the share of girl's pay relative up to date guy's pay on the subindustry level in each kingdom (measured on a ratio scale, as a percent). Previous students have up-to-date for a research in this up-to-date using a quantitative studies method [10,4]. A higher conclusion of statistical evaluation up to date examine facts among the studies and variables may boom analytical and generalized effectiveness making use of a quantitative research method. The statistics evaluation furnished facts approximately gender-associated pay fairness in all 50 states plus the District of Columbia and the industries inside each nation.

Analyses of gender-related pay facts were performed with IBM SPSS records software up-to-date gender-related pay equity specifics. The U.S. Census Bureau administered the 2017 American network Survey and this dataset furnished a large quantity of gender-associated pay records for men and women at an individual country degree. Gender-related pay information retrieved from the united states Census Bureau (American network Survey) have been analyzed up to date the gender-associated pay equity percentage between ladies and men. The yankee network Survey raw information furnished the percentage of gender-associated pay equity for each subindustry inside every kingdom. On this have a look at, the percentage of gender-associated pay represents the repayment ratio among women and men. Better probabilities indicated higher pay equity for ladies relative updated guys. Values underneath 100% indicated that women had been paid less than men, and percentages above 100% indicated that ladies were paid extra than men. States had been damaged in updated the 5 important industries up to date updated industry bias. The 2017 American network Survey raw dataset furnished all records up-to-date (probabilities).

The variables were gender-related pay equity, kingdom, and enterprise. It is vital up to date note that the based variable of gender-related pay equity was utilized for each research questions. The analysis concerned two ANOVA exams evaluating states and industries in phrases of the established variable.

A Bonferroni correction up-to-date applied up to date the level of statistical significance updated save your inflation of type I blunders. The corrected alpha has become .025 ($.05/2 = .025$). the following studies questions and corresponding hypotheses guided this quantitative comparative look at:

RQ1: Are there any statistically widespread differences in gender-associated pay equity the various 51 states (up-to-date D.C.) in the united states of America public zone?

H10: There's no statistically considerable difference in gender-associated pay equity many of the 51 states (up-to-date D.C.) in the United States public region.

H1a: There's at least one statistically big difference in gender-associated pay fairness the various 51 states (up-to-date D.C.) in the America public sector.

RQ2: Are there any statistically great variations in gender-associated pay equity a number of the five predominant industries within the America public sector?

H20: There's no statistically substantial distinction in gender-related pay fairness among the five major industries within the United States public region.

H2a: There is as a minimum one statistically giant difference in gender-associated pay equity a few of the 5 most important industries inside the Unit ed States public quarter.

Populace and sample choice:

This look at popular populace became 2.1 million respondents up-to-date the American Community Survey of women and men from the united states public zone. All 50 states, up to date the District of Columbia, had been covered. moreover, the 5 principal industries in each country have been covered.

The original 2.1 million facts up-to-date had been archival data from the USA Census Bureau statistics authorization up-to-date obtained from the us Census Bureau, middle for financial research (CES), which affords public-use records.

electronic mail confirmation of approval and authorization up-to-date the USA Census Bureau facts turned in updated completed (see Appendix B.). If authorization from America Census Bureau had now not been acquired, other public archival databases would have been up-to-date.

Addressing the hassle declaration and answering each gender-associated pay fairness studies questions required two one-way ANOVAs of aggregated archival information accrued from America Census Bureau records (American community Survey). The aggregated archival was retrieved from the 2017 ACS dataset. Uncooked statistics were aggregated updated offer a unmarried facts for every of the 36 subindustries from each of the fifty one states (including D.C.), representing the 1,834 records up-to-date (District of Columbia lacked facts from two subindustries).

The unit of remark in the survey was the man or woman respondent and the unit of evaluation subindustry identified through nation, applied for analysis. The unit of evaluation had twin identity: subindustry X country (36 subindustries X 51 states = 1836 instances). Since the unit of analysis had dual identification, pay facts (unit of evaluation) from the 36 subindustries represent the dataset for each questions.

The ACS facts are archival and publicly accessible, indicating that no player approvals have been wished. Sample and target sizes were the identical up-to-date the archive consists of up to total of 2.1 million information. The original 2017 ACS dataset protected 3,526,808 responses.

The final data set consisted of 2,145,639 statistics up to daters updated up-to-date the USA Census Bureau only accepts fully completed surveys. From the authentic 2.1 million data up-to-date, statistics were in comparison up to date updated 1,834 data up-to-date for analyses (i.e., 51 states (up-to-date D.C.) and the 36 subindustries inside each kingdom). District of Columbia lacked information from subindustries, ensuing in up-to-date dataset of 1,834 as compared up-to-date, 836. The two subindustries not represented inside the District of Columbia are within the fundamental enterprise of herbal resources, production and renovation. The 2 subindustries are farming, fishing, forestry, and construction and extraction occupations.

The USA Census Bureau has a minimal popular self-assurance stage of 90%, with the margin of blunders (MOE) = $1.645 \times \text{S.E.}$. S.E. stands for fashionable error (S.E.) the foundational measure of the range of an estimate due up updated sampling. The Census Bureau states exchange confidence ranges in statistics 95% and 99%, $\text{MOE} = 1.96 \times \text{S.E.}$, and $2.58 \times \text{S.E.}$ accomplishing the best level of confidence in data is important; utilizing a bigger geographical length and mixing estimates throughout traits and geographies lowers the risk of estimate sampling variability [22]. This researcher utilized a massive geographic size up to date ether with all characteristics updated achieve 99% confidence in facts integrity (supplementary Table 1).

Instrumentation:

The source of all data used on this look at up to date the U.S. Census Bureau, which accumulated the statistics via the American Community Survey (ACS). The 2017 ACS accrued data from 2.1 million public quarter personnel. The us Census Bureau collects facts thru survey techniques: online and paper. The united states Census Bureau seeks up-to-date achieve a full-size majority of information gathered through the internet site, on-line survey, and a mail-in alternative. The U.S. Census Bureau results are required under regulation thirteen, U.S. Code, Sections 141, 193, 221, and inform how 675 billion dollars of federal bucks are dispersed throughout the united states. The distribution of 675 billion greenbacks is 29% of all USA federal assist.

Information evaluation:

Two one-way ANOVA analyses had been used updated examine gender-associated pay fairness across states and industries. IBM SPSS information software changed in updated used for the complete evaluation.

Preceding researchers have analyzed incomplete data that did now not encompass country wide records broken down via country, requiring a bigger, extra numerous pattern size [18-21].

The minimal pattern size for this quantitative, no experimental, comparative research changed in updated envisioned in G*power 3.1.9.7. for 2 one-manner ANOVAs (fixed effects, omnibus, one-way) with the equal established variable and wonderful, unbiased variables. The enter covered a predicted medium impact size ($f = .25$), corrected alpha (.025), minimum power .95 (that means 5% risk of type II blunders), and the maximum number of as compared groups (51).

The anticipated minimum pattern up to date 816 instances (gender-related pay ratio among ladies and men) with entire facts for every studies question (see Appendix F). The researcher brought that 15% (123 instances) up to date discard outliers or use nonparametric tests in case of unresolved assumption violations for the favored parametric analysis. This raised the minimal pattern size updated 939 cases.

The very last pattern included 1,834 cases (gender-associated pay ratio between males and females) and exceeded the minimal sample length for both studies questions.

IBM SPSS 25 up-to-date statistics have been executed with a multi-step system.

1. Country information had been downloaded from the USA Census and American Community Survey databases in a CSV format that is importable up-to-date IBM SPSS up-to-date records software.

2. Each country, the District of Columbia, and industry up to date assigned various representing evaluation for SPSS

3. A nation and the District of Columbia information point covered 36 person industries inside every country and the District of Columbia.

4. State and the District of Columbia records were calculated up to date updated up-to-date dataset of one,834 statistics points for analyses (51 states, updated D.C.) and the 36 subindustries within every nation).

5. A CSV document up-to-date organized up to date symbolize the gender-related pay equity for 1,834 records up to date for analyses (51 states, up to date ether with D.C.) and the 36 subindustries within each state).

6. The five most important industries are defined from the authentic 36 subindustries. Archival facts provide extensive enterprise facts update.

7. The CSV document up to date updated imported in updated IBM SPSS 25 up-to-date records software program.

8. The analysis system compared the way through the gender-related pay fairness price take a look at variable of the 1,834 records up-to-date.

9. With the collection of nation records in up-to-date IBM SPSS format, a CSV record up-to-date imported in up to date IBM SPSS 25 up to date rate updated facts software program.

10. The six assumptions for one-manner ANOVA have been tested the usage of the Shapiro-Wilk check, Levene's check, and Kolmogorov-Smirnov check.

11. If the sample failed assumptions, then a nonparametric test up-to-date be executed for both units of impartial organizations.

12. IBM SPSS univariate options were set up to date descriptive information and homogeneity of variance test.

13. The extent of statistical significance became corrected up-to-date half up-to-date mitigate inflation of kind I error.

14. The only-manner ANOVA analysis method sought up to date updated facts interaction utilizing a preferred linear version and univariate evaluation.

15. The univariate analysis variables defined a based variable of fee (gender-related pay fairness), country, and industry up to date updated.

16. IBM SPSS turned in updated used up to date updated the statistics.

Results:

On this non-experimental comparative look at, the researcher as compared facts accrued from the USA Census, American community Survey, 2017. The dataset enabled the researcher updated study a national sample. The non-experimental comparative method provided the research framework updated examine gender-associated pay fairness among men and women of all 50 states, the District of Columbia, and 5 significate industries (36 subindustries).

The structured variable up to date gender-related pay equity. This variable became operationalized as the percentage of ladies pay relative up to date men's pay at the subindustry stage, through country. Similarly, the sample of facts is in keeping with individual country records in the USA.

The descriptive facts summaries of the gender-related pay fairness are shown in Tables 1 and a couple of. The suggest gender-associated pay equity is 74.73% (SD = 14.24%) for the overall populace.

The highest gender-related pay equity within the dataset is 118.sixteen%, even as up-to-date gender-associated pay equity of the general populace is 32.71%, proven in Tables 2 and a couple of. It up to date be referred updated that male samples have higher pay information than woman samples up to date the share is below a hundred% which means that men are in want of women (ladies are paid less than guys).

Consequently, a gender-associated pay equity difficulty up to date observed; but, the importance of the difference of the gender-related pay equity up-to-date decided using a one-manner ANOVA updated determine an enormous distinction in gender-associated pay fairness through states and enterprise of the united states public area (supplymentary table 2).

Up to date the composite abuse scale revised (CASR-SF).

Composite abuse scale revised refers up-to-date the relationship we mean a current companion. A major interpretation of CASR-SF is up-to-date validate the reliable brief of self-reporting size developed at the same time as utilizing a mixed-approach method. Most of the people cognizance of CASR-SF is up to date cognizance at the severity and depth of the records point captured (supplementary table 3).

The outcomes presented in table 4 imply a statistically huge difference across the 51 states (up to date D.C.) in terms of gender-related pay fairness, $F(50, 1740) = 1.69$, $p = 0.019$. The difference is vast up-to-date the p-cost is below the corrected stage of importance fee ($\alpha = .025$). No checks were achieved due updated the very massive variety of compared organizations (51). based updated on those findings, which showed statistically sizeable differences in gender-associated pay equity throughout the 51 states (which include D.C.) for U.S. public sector personnel, the null hypothesis for studies query 1 became rejected (supplementary table 4).

States with the lowest gender-related pay equity, women were paid lowest to men (measured as percent difference between the median values for males and for females) were Idaho ($M = 66.91\%$; $SD = 16.55\%$), Utah ($M = 67.53\%$; $SD = 15.35\%$), Louisiana ($M = 69.53\%$; $SD = 12.66\%$), Wyoming ($M = 70.80\%$; $SD = 18.93\%$), and Connecticut ($M = 71.58\%$; $SD = 13.31\%$). The states with the highest (i.e., best) gender-related pay equity were the District of Columbia ($M = 81.69\%$; $SD = 13.31\%$), Nevada ($M = 81.18\%$; $SD = 73.86\%$), Arizona ($M = 80.67\%$; $SD = 11.75\%$), Vermont ($M = 79.55\%$; $SD = 17.82\%$), and Maryland ($M = 79.15\%$; $SD = 14.17\%$).

The categorical unbiased variable defined five businesses that had been as compared: (1) management, commercial enterprise, technological know-how, and art occupations, (2) provider occupations, (3) sales and workplace occupations, (4) natural assets, construction and renovation occupations, and (5) manufacturing occupations.

A degree of importance of .1/2 up to date updated also used within the one-way ANOVA. the only-manner ANOVA consequences determined the importance of the difference in gender-related pay fairness by using enterprise; the consequences are proven in desk four.

The only-manner ANOVA discovered a huge difference within the gender-associated pay equity many of the five major industries, $F(4, 1735) = 17.00$, $p < .001$ (table 4). There is a good sized distinction because the p-cost up-to- date the F statistic is lower than the corrected level of significance ($\alpha = .05$).

It's far essential updated observe that evaluation of the main industries had 51 fewer data up-to-date because "civilian employed populace sixteen years and over with earnings" does not have a selected industry since it represents all data updated that had been no longer especially categorized underneath one of the 5 most important industries.

The test results of the video Games-Howell assessments (table 5) diagnosed the statistically great differences for more than one pairings of organizations. Mainly, there had been significant differences within the gender-associated pay equity between management, business, science, and artwork occupations; and natural assets, creation, and upkeep career ($p < 0.001$) by way of a mean difference of 4.34%. There were tremendous differences in the gender-related pay equity between control, business, technological know-how, and art occupations; and production occupations ($p < 0.001$) by means of a mean difference of 8.35%.

There were massive differences within the gender-related pay fairness between service occupations; and natural resources, creation, and protection profession ($p = 0.01$) by way of a median difference of 4.14%.

There have been massive variations in the gender-associated pay fairness between provider occupations and production occupations ($p = 0.001$) by means of a median distinction of 4.75% also, there was a significant difference in the gender-related pay equity between income and workplace occupations and production occupations ($p < 0.001$) by way of a median distinction of 5.43%. There was a great distinction within the gender-related pay equity between natural sources, production, and maintenance occupations, and production occupations ($p = 0.04$) by way of a mean distinction of 4.01% (supplementary table 5).

The assessment of means in table 6 display that the industries with the very best gender-associated pay equity (i.e., up-to-date percent of woman profits relative up to date male income) had been production occupations ($M = 68.18\%$; $SD = 11.84\%$); and natural resources, production, and upkeep occupations ($M = 72.18\%$; $SD = 16.53\%$) (supplementary table6).

The industries with the lowest gender-associated pay equity (i.e., maximum percentage of lady profits relative updated male income) were management, business, technological know-how, and art occupations ($M = 76.53\%$; $SD = 12.29\%$) followed through carrier occupations ($M = 76,33\%$; $SD = 14.95\%$).

Those outcomes provided evidence of statistically great variations in gender-related pay fairness throughout the 5 principal industries for the USA public quarter up to date on these findings of the only-manner ANOVA, the null speculation for research question 2 became rejected (supplementary table 7).

Discussion:

The observe consequences have profound sensible implications and programs concerning gender-related pay equity. The take a look at outcomes advocate that gender-associated pay fairness is extensively prompted with the aid of the individual state wherein a woman lives and enterprise.

As said, before the negative results of gender-related pay equity are felt at some stage in ours. Pay fairness is up-to-date that impacts 74.6 million women people inside the civilian hard work force [13]. Same pay among ladies and men would reduce poverty for working girls from 8.2 up-to-date 4 percent. Every of the 50 person states would benefit from a growth of funds in up-to-date their economies [14]. Every other terrible impact of gender salary inequality is that gender-related pay fairness contributes up to date elevated charges of anxiety and depression amongst women [15]. In a partnership with the middle for administrative center intellectual fitness, the yank Psychiatric association expressed big subject on updated of gender-related pay fairness. The President of the yankee Psychiatric affiliation, Renee Binder, MD, said that gender-related pay equity is extra impactful than financial problems and contributes updated intellectual issues [14]. Researchers have proven that income inequality increases the threat of obesity and coronary heart attack [17].

Scholars have suggested that gender-related pay fairness is appreciably primarily based on the state of the USA; therefore, greater sources can be carried out in updated regions of need. Such resources can consist of money, economic strain, social stress and schooling because gender-associated pay equity has substantial un-favorable fitness outcomes, there is a want updated boom the international locations intellectual and bodily fitness assets. Federal funding may be withheld from states for not following federal laws.

Destiny Implications:

The consequences of locating out that states have not adhered up to date preceding rules will create up to questions. Negative findings up to date motive a halt of billions of dollars of federal price range.

Federal investment is meant updated help the kingdom in infrastructure, education, relief, and financial advantages.

This is a completed dataset up to date updated offer sensible applications because the blueprint of gender- associated pay fairness, proving people' information updated law and political strain for alternate and advantage 74.6 million American girls people. States of the USA aren't shielding the ladies that live within them.

The mental, physical, social, and financial fitness of those women is negatively effective by way of gender-related pay fairness. It has been discovered that individual states aren't protecting women employees. the failings of country governments, corporations, and companies cannot go disregarded. Destiny researchers have supplied the framework for assessing why their kingdom affects gender-associated pay fairness. More country and industry-particular research may be carried out up to date gender-related pay fairness on the country and enterprise degrees.

Conclusion:

This recent study identified 11 variants first reported in Saudi Arabia. Seven new variants of the HBB gene were observed in the study population. The impact of these new variants on the phenotypic properties and actual function of β -globin is currently unknown and needs to be investigated. The identified β -globin mutations could potentially be used as genetics in the research area.

Declaration of Conflict of interest:

The author declares no conflict of interest.

References:

1. Wade M, Fiorentino S (2017) Gender pay inequality: An examination of the Lilly Ledbetter Fair Pay Act six years later. *Advancing Women in Leadership J.* 37:a11.
2. Adams JS (1965) Inequity in social exchange. *Advances in Experimental Social Psychology.* Sci Rec. 2:267-299.
3. Dennis GA (2016) Gender pay disparity among women. *Exceptional Children.* 85(2):124.
4. Rosado OM (2018) Trends in gender wage gaps: A qualitative study. Doctoral dissertation, Northcentral University. 18:10815703.

5. Smit BW, Montag-Smit T (2019) The pay transparency dilemma: Development and validation of the pay information exchange preferences scale. *J of Applied Psych.* 104(4):537-558.

6. White KK (2019) The effect of union affiliation on salary for female community college faculty. *Saint Louis Pro Disser Publis.* 19:2766.

7. Phillips AN (2018) Promulgating parity: An argument for a states-based approach to valuing women's work and ensuring pay equity in the united states. *Tulane Law Review.* 92(3):719-743.

8. Geoghegan FJ (2018) The gender wage gap: How paid family leave affects women of childbearing age (Master's thesis, Georgetown University). *ProQuest Dissertations & Theses Global.* 51:1050853.

9. Burn I (2018) Not all laws are created equal: Legal differences in state non-discrimination laws and the impact of LGBT employment protections. *J of Labor Res.* 39(4):462-497.

10. Swain P (2019) The lived experiences of U.S. female employees who have experienced gender-based wage disparity (Doctoral dissertation, Georgia State University). *ProQuest Dissertations & Theses.* 19:27667298.

11. Lobel O (2020) Knowledge pays: Reversing information flows and the future of pay equity. *Columbia Law Review.* 120(3):547.

12. Schulze E (2018) Closing the gender pay gap could have big economic benefits. *CNBC.*

13. DeWolf M (2017) 12 stats about working women. *U.S. Department of Labor Blog.*

14. Status of Women (2020) The economic impact of equal pay by state.

15. Platt J, Prins S, Bates L et al. (2016) Unequal depression for equal work? How the wage gap explains gendered disparities in mood disorders. *Social Science & Med.* 149:1-8.

16. Kuber R (2016) Gender pay gap contributes to increased rates of depression and anxiety among women. *Workplace Mental Health.*

17. Pabayo R, Fuller D, Lee E et al. (2018) State-level income inequality and meeting physical activity guidelines: Differential associations among U.S. men and women. *J of Public Health.* 40(2):229-236.

18. Obloj T, Zenger T (2020) The influence of pay transparency on inequity, inequality and the performance-basis of pay in organizations. *SSRN Electronic J.* 6:646-655.

19. Cortés P, Pan J (2019) When time binds: Substitutes for household production, returns to working long hours and the skilled gender wage gap. *J of Labor Econ.* 37(2):351-398.

20. Goldin C (2017) The expanding gender earnings gap. *A Econ Review.* 107(5):110-114.

21. Blau FD, Kahn LM (2017) The gender wage gap: Extent, trends and explanations. *J of Econ Lite.* 55(3):789-865.

22. Fuller S (2018) Using American community survey estimates and margins of error *U.S. Census Bureau, Decennial Statistical Studies Division.*

Supplymentary tables:

Confidence Level	Margin of Error (MOE)	MOE For Example Estimate
90%	1.45 x S.E.	+/-3,778
95%	1.96 x S.E.	+/-4,501
99%	2.58 x S.E.	+/-5,925

Table 1: United States Census Bureau Confidence Chart.

Median earnings (%)	N	Minimum	Maximum	M	SD
Gender -Related Pay Equity	1791	32.71%	118.16%	74.73%	14.24%

Table 2: Descriptive Statistics Summaries of Gender-Related Pay Equity Data.

CASR-SF	Mean	Median	Standard Deviation	Standard Error	z-Skewness	z-Kurtosis
Median Earnings	74.73%	74.71%	14.24%	0.34%	-0.02	0.13
RQ1: Are there any statistically significant differences in gender-Related pay equity among the 51 states(including D.C.) in the United States public sector?						

Table 3: Descriptive Statistics for Gender-Related Pay Equity Variables of Measured as CASR-SF (N = 1,791).

	Sum of Squares	df	Mean Square	F	P
Between Groups	16,864.70	50	337.29	1.69	0.019*
Within groups	3,46,794.85	1740	199.3		
Total	3,63,659.56	11790			

Table 4: Results of One-Way ANOVA for Gender-related Pay Equity by State.

	Sum of Squares	df	Mean Square	F	p
Between Groups	13,660.15	4	3,415.03	17	<0.000*
Within Groups	3,48,449.65	1735	200.83		
Total	3,62,109.80	1739			
*Significant difference at the level of significance of 0.025					

Table 5: Results of the One-Way ANOVA for Gender-Related Pay Equity by Major Industry.

(I) Industry	(J) Industry	Mean Difference (I-J)	S.E.	Sig.**	95% Confidence Interval	
					Lower Bound	Upper Bound
1	2	0.20%	0.87%	1	-2.18%	2.58%
	3	2.91%	1.25%	0.14	-0.50%	6.33%
	4	4.34%	1.16%	0.00**	1.18%	7.50%
	5	8.35%	1.11%	0.00**	5.31%	11.39%
2	3	2.72%	1.35%	0.26	-0.97%	6.41%
	4	4.14%	1.27%	0.01**	0.68%	7.60%
	5	8.15%	1.23%	0.00**	4.80%	11.50%
3	4	1.43%	1.55%	0.89	-2.81%	5.66%
	5	5.43%	1.52%	0.00**	1.29%	9.58%
4	5	4.01%	1.44%	0.04**	0.06%	7.95%
*Industries: (1) management, business, science, and art occupations (2) service occupations (3) sales and office occupations (4) natural resources, construction, and maintenance occupations and (5) production occupations.						
**The mean difference is significant at the 0.05 level.						

Table 6: Results of the Games-Howell Test for Gender-related Pay Equity by Major Industry*.

Main Industry*	N	M (%)	S.D. (%)	Min (%)	Max (%)	z-Skewness	z-Kurtosis
1	810	76.53	12.29	37.55	116.33	-0.07	0.52
2	391	76.33	14.95	32.71	116.42	0	0.07
3	153	73.61	20.01	37.03	111.03	-0.15	-1.22
4	184	72.18	16.53	35.83	118.16	0.25	0.01
5	202	68.18	11.84	36.92	114.34	0.37	0.72
Total	1740	74.8	14.43	32.71	118.16	-0.03	0.06
*Note: Main industry classification: (1) management, business, science, and art occupations; (2) service occupations; (3) sales and office occupations; (4) natural resources, construction, and maintenance occupations; and (5) production occupations.							

Table 7: Results of the Games-Howell Test for Gender-related Pay Equity by Major Industry*.